

DEYUNG SUN

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Mechanical Engineering M.S. candidate (GPA: 3.75/4.0) with hands-on experience in **robotic hardware design, ROS 2 system integration, and experimental validation**. Experienced in sensor fusion (LiDAR, IMU, odometry), autonomous navigation, and deploying real-world robotic systems. Proficient in Python, C++, MATLAB, and Linux. Bilingual in English and Mandarin.

EDUCATION

M.S. Mechanical Engineering — Cal Poly Pomona

Exp. 2026 | GPA: 3.75 / 4.0

Coursework: Fundamentals of Robotics, Robot Motion Planning, Machine Learning, Control of Mechanical Systems, Systems & Signals, Numerical Methods

B.S. Mechanical Engineering — National Cheng Kung University, Taiwan

Jun 2023

EXPERIENCE

Graduate Research Assistant — Autonomous Vehicle Laboratory | Cal Poly Pomona

Jun 2025 – Present

- Designed and integrated **mechatronic subsystems** (steering mechanisms, sensor mounts) for a campus autonomous driving robot, cutting prototype iteration time by ~30% through structured CAD-to-fabrication workflows
- Fused **LiDAR, IMU, and wheel odometry** data across ROS 2 nodes, improving localization consistency and reducing dead-reckoning drift in campus navigation tests
- Built and executed **repeatable test scenarios** in Gazebo and physical environments, validating navigation stack performance across 10+ scenario configurations
- Applied **Python/MATLAB data analysis** to tune Nav2 planner and controller parameters, achieving measurable improvement in obstacle avoidance reliability
- Documented test procedures, failure modes, and corrective actions; maintained structured logs that accelerated root cause identification across mechanical, electrical, and software interfaces
- Collaborated cross-functionally to resolve system integration issues, reducing bring-up time for new hardware configurations

Engineer | Proaction Technology Inc., Taiwan

Jul 2023 – Dec 2023

- Deployed, maintained, and troubleshoot **electromechanical monitoring systems** across multiple field sites, achieving consistent uptime under operational constraints
- Diagnosed and resolved failures in **sensors, wiring, and mechanical components**, restoring system functionality through structured debugging and verification procedures
- Executed functional testing, safety checks, and commissioning for newly installed units; documented repair records to build a reliability knowledge base

PROJECTS

Autonomous Navigation System with Dynamic Obstacle Handling | ROS 2 + Gazebo

2024 – Present

- Developed a full autonomy stack inspired by Nav2, implementing **A***, **BFS global planners** and a **custom DWA local planner** for real-time trajectory control with dynamic replanning
- Designed **LiDAR-driven occupancy grid pipelines** and closed-loop perception–planning–control architecture; validated performance across different simulated environments
- Modeled robot in **URDF/SDF/Xacro**, integrated `ros_gz_bridge`, and verified coordinate frames via RViz TF debugging

Logistics Warehouse Robot — Competition | LEGO EV3 + PID Control

2022

- Designed drivetrain and gripper mechanism, implemented **PID control**, and iteratively tuned based on live test data — earned **Top 3 in the department**

TECHNICAL SKILLS

Robotics & Autonomy: ROS 2 (Jazzy), Gazebo, Nav2, sensor fusion (LiDAR / IMU / odometry), URDF/SDF, RViz, TF, mobile robot integration

Programming: Python, C++, MATLAB, Linux (Ubuntu), Git, Bash

CAD & Fabrication: SolidWorks, AutoCAD, FDM 3D printing (Bambu Lab), manual milling & lathe, 5-axis CNC exposure, GD&T basics, DFM/DFA

Testing & Analysis: Functional & acceptance testing, Gazebo simulation, failure analysis, data-driven tuning, technical documentation

Languages: English (fluent), Mandarin (native)